

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

Coursework: Systems for Data Science, Performance Engineering, Advanced Machine Learning, Optimization in CS, Advanced NLP

September 2025 – May 2027 (Expected)

Amherst, MA

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, Java, C++, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, microservices, async services, integration testing, authentication, rate limiting, audit logging

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Databases & Web

MongoDB, PostgreSQL, Redis, SQLite, React, Electron, Webpack, GraphQL

AI-Assisted Dev & GenAI

Claude / Claude Code, GitHub Copilot, Cursor, ChatGPT / Codex, OpenAI / Groq APIs, RAG, LLM agents, PyTorch, Hugging-Face Transformers

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Owned end-to-end backend for **multi-tenant NHS eTriage platform** serving **8+ UK clinical clients** and processing **10,000+ monthly submissions** (Node.js, MongoDB), integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Architected and shipped a full-stack **data annotation platform** (React, Node.js, MongoDB) used daily by the AI team across the full design-through-deployment lifecycle, replacing spreadsheet workflows with secure role-based REST APIs and cutting manual operations by **100+ hours/month** while improving labeling throughput by **60%**.
- Drove operational excellence across the eTriage stack: monitored production logs during feature releases and client integrations, troubleshooted and resolved tenant-specific production incidents, hardened authentication and rate limiting, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect sensitive clinical data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ production RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes consumed daily by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that converts open-ended topic queries into structured entity tables via typed retrieval planning, page rerank, deterministic extraction with LLM fallback, output verification, and **cell-level provenance** for every result.
- Hardened the service with **150+ automated tests, provider routing and fallback** across OpenAI/Groq backends, and pipeline instrumentation, leveraging GenAI tools (Claude Code, Cursor) as engineering accelerators while keeping system behavior grounded in deterministic validation and a reviewer-facing trust UI.

Spark ETL Performance Optimization - NYC TLC Taxi Dataset

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a distributed Spark ETL pipeline on the NYC TLC taxi dataset by tuning broadcast joins, caching, partitioning, adaptive Spark settings, and column pruning to reduce shuffle volume and wall-clock runtime.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks and documented a repeatable optimization workflow comparing naive vs. tuned runs as a reusable reference for downstream analytics jobs.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built a cloud-native AI research assistant on **Cloudflare Workers**: agent fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time.
- Designed durable workflow boundaries so multi-step research tasks survive process restarts and individual tool failures; agent state and conversation history persist server-side in D1 without an external database, demonstrating fault-tolerant cloud-native design.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.